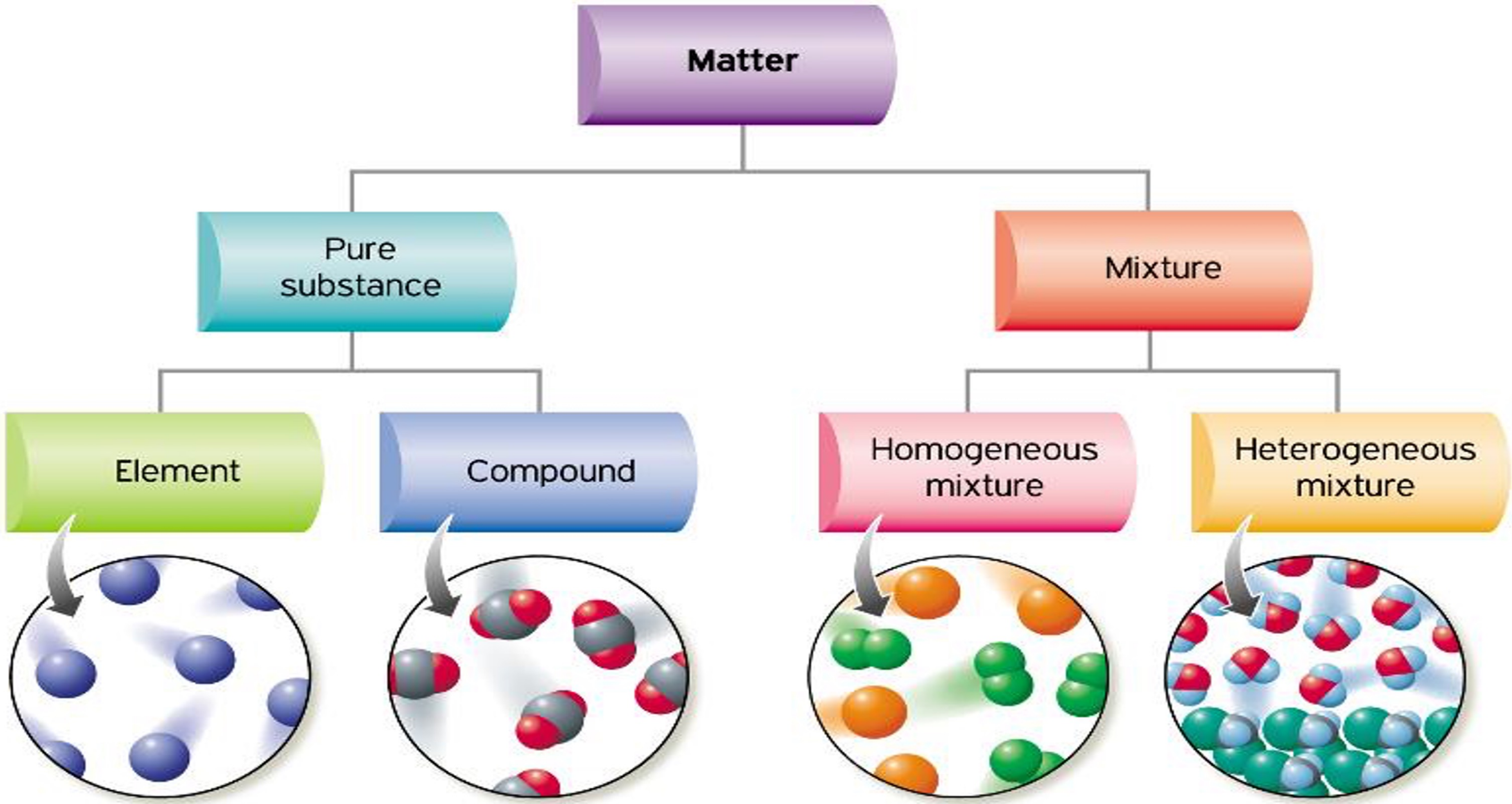
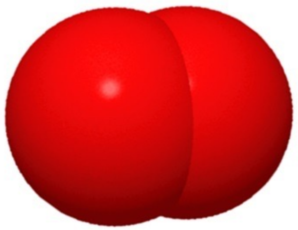
MIXTURES AND SOLUTIONS

Organization of Matter

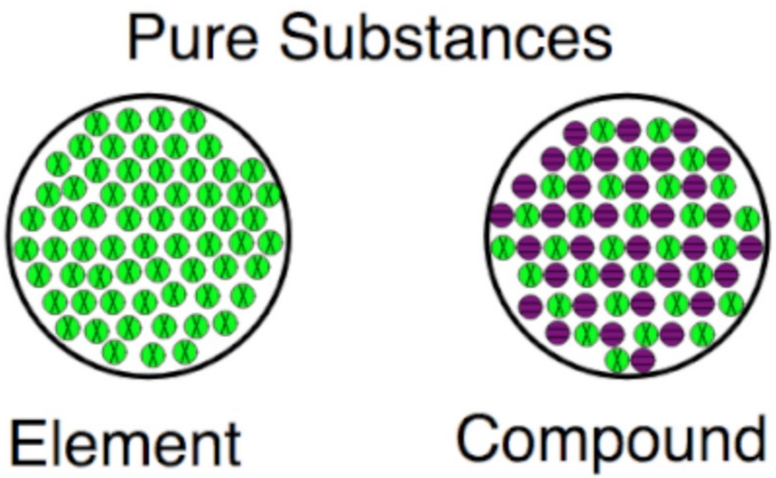


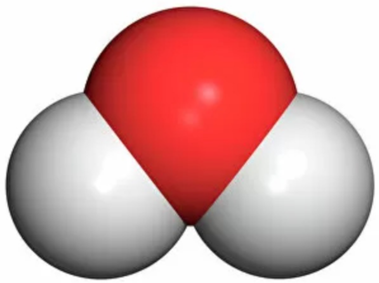
Pure substances

- made up of only one kind of particle and have a fixed structure.
- *Cannot* be separated by physical means (physical change).
- There are two types of pure substances, **elements** and **compounds**.



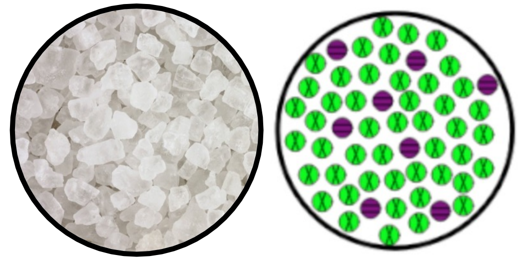
O_2
Molecular
Oxygen





H_2O
Water
Molecule

Homogeneous mixture (solution)

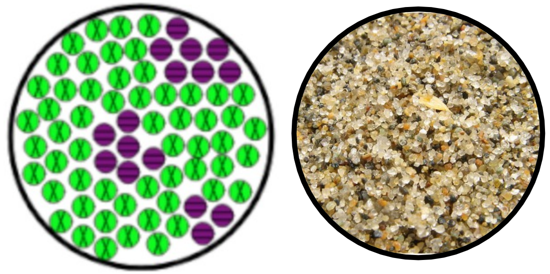


Homogeneous

- Uniform composition = the same all throughout
- “Appear” to contain only one type of substance.
- You cannot see the parts that make up the substance.
- Particles are < 1 nm in size.
  - Examples: air, salt water

Heterogeneous mixture (mixture, suspension, colloid)

- Medium/large particles (larger than 1 nm)
- Composition is not uniform throughout
- You can visibly see the 2 or more parts.
  - Examples: pizza, wood, chocolate chip ice cream

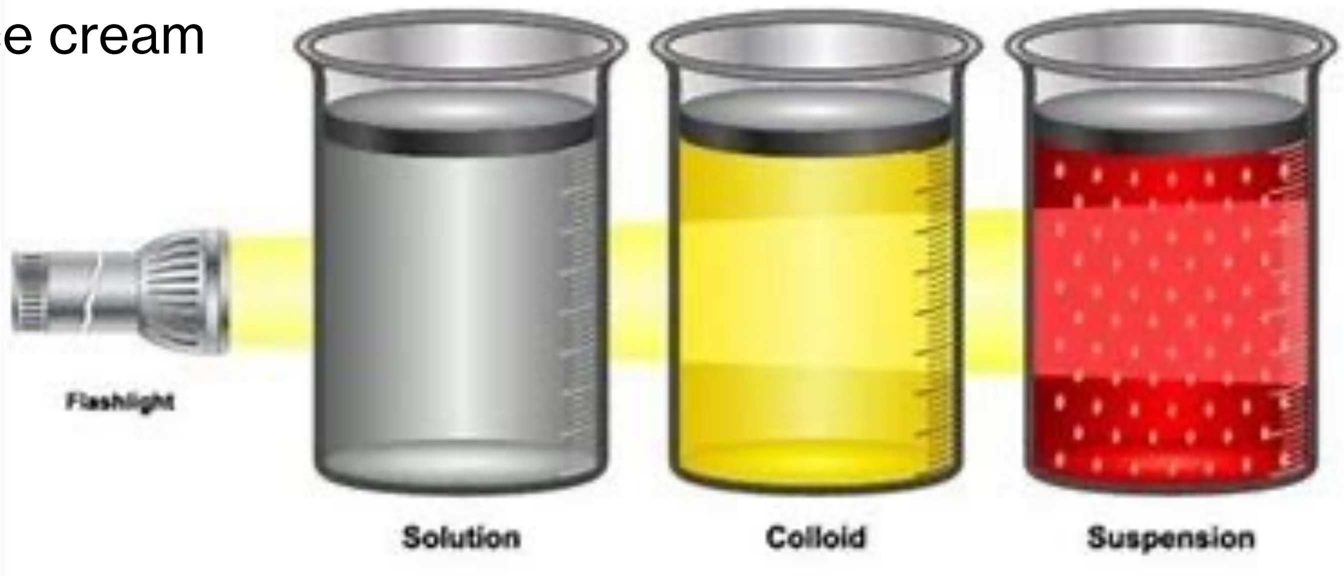


Heterogeneous

Suspensions & Colloids

Suspension- particles will settle out if agitation stops.
larger particles: > 1000 nm, visible in the light
Examples: tomato/orange juice.

Colloid- particles will stay suspended for a long time
medium-sizes particles :1-1000 nm.
Examples: fog, gelatin, Jello, face cream



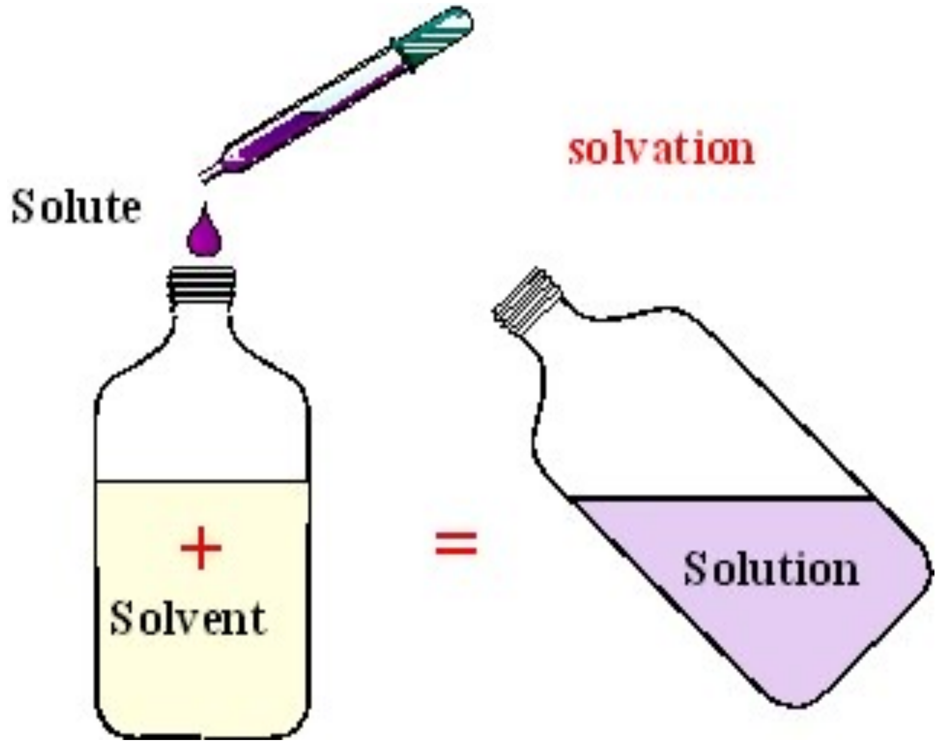
Sometimes colloids are considered **homogeneous**, depending on the scale of interest.

solution - a homogeneous mixture in the liquid phase.

- **Solute:** the substance that <u>is dissolved</u> in another substance

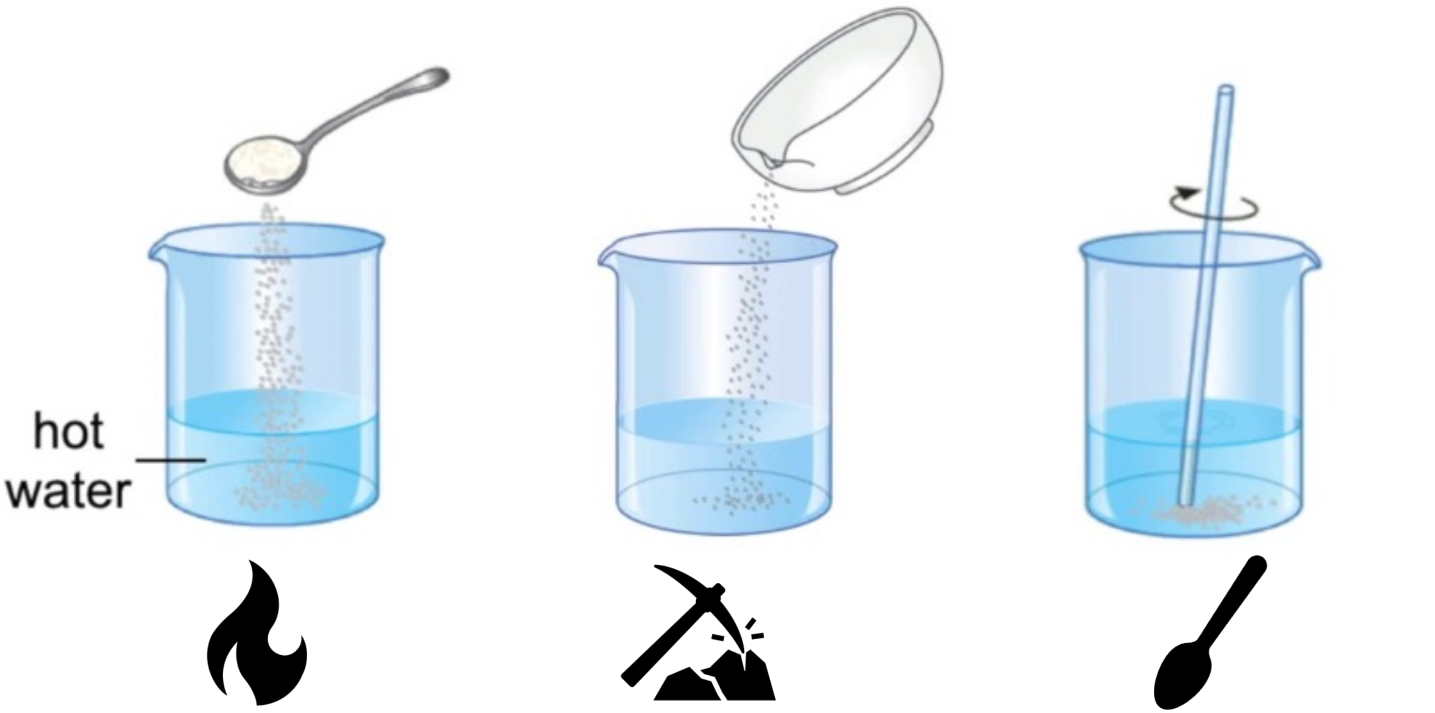
- **Solvent**: the substance that something is dissolved into (the dissolver)

The solute is in the smaller proportion, while solvent is the material in the larger proportion



How to Dissolve Faster

Solubility- the ability of a substance to dissolve in another substance.



Add Heat   Smaller Particles   Agitation (stirring)

Sieving

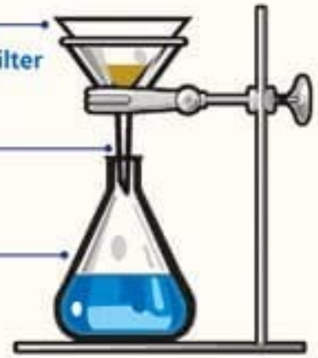
Can separate large particles from small particles.

Sieve

- The sand and rock mixture is shaken through.
- The small holes in the sieve let the sand particles through and prevent the larger rocks from passing through.

Filtration

Can separate solids that are insoluble from a liquid.



Filter Paper

Sand gathers in the filter paper. Water passes through.

Funnel

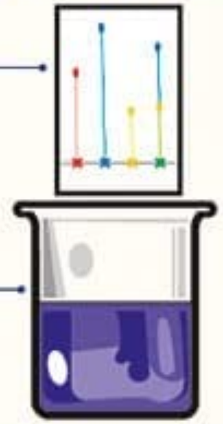
Channels the water into the flask.

Flask

Gathers the water after filtration.

Chromatography

Can separate different colour dyes.



Paper

- Spots of ink are placed on a pencil line.
- Solvent soaks up to the ink and each colour seeps upwards at a different rate.

Solvent

The paper is placed in the solvent which soaks up to the ink spots.

Evaporation

Can separate solids that are soluble from a liquid.

Salt Water

Evaporating Dish

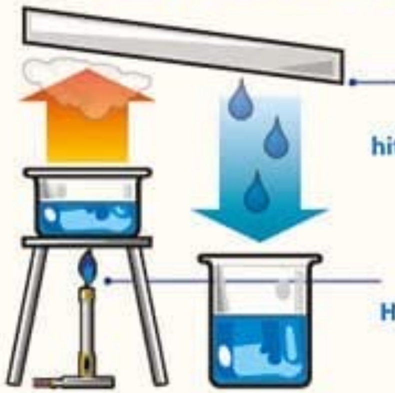
Water evaporates into the air, leaving only the salt.

Bunsen Burner

Heats the salt water so that it boils and turns to steam.

Condensation

Condensing is when water vapour changes into liquid water.



Cold Surface

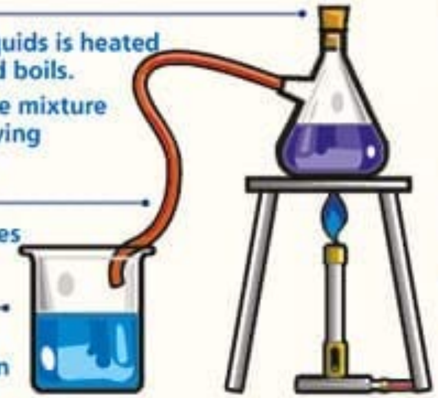
The steam hits the surface and condenses into water droplets.

Bunsen Burner

Heats the water so that it boils and turns to steam.

Distillation

Can separate a solvent from a solution.



Flask

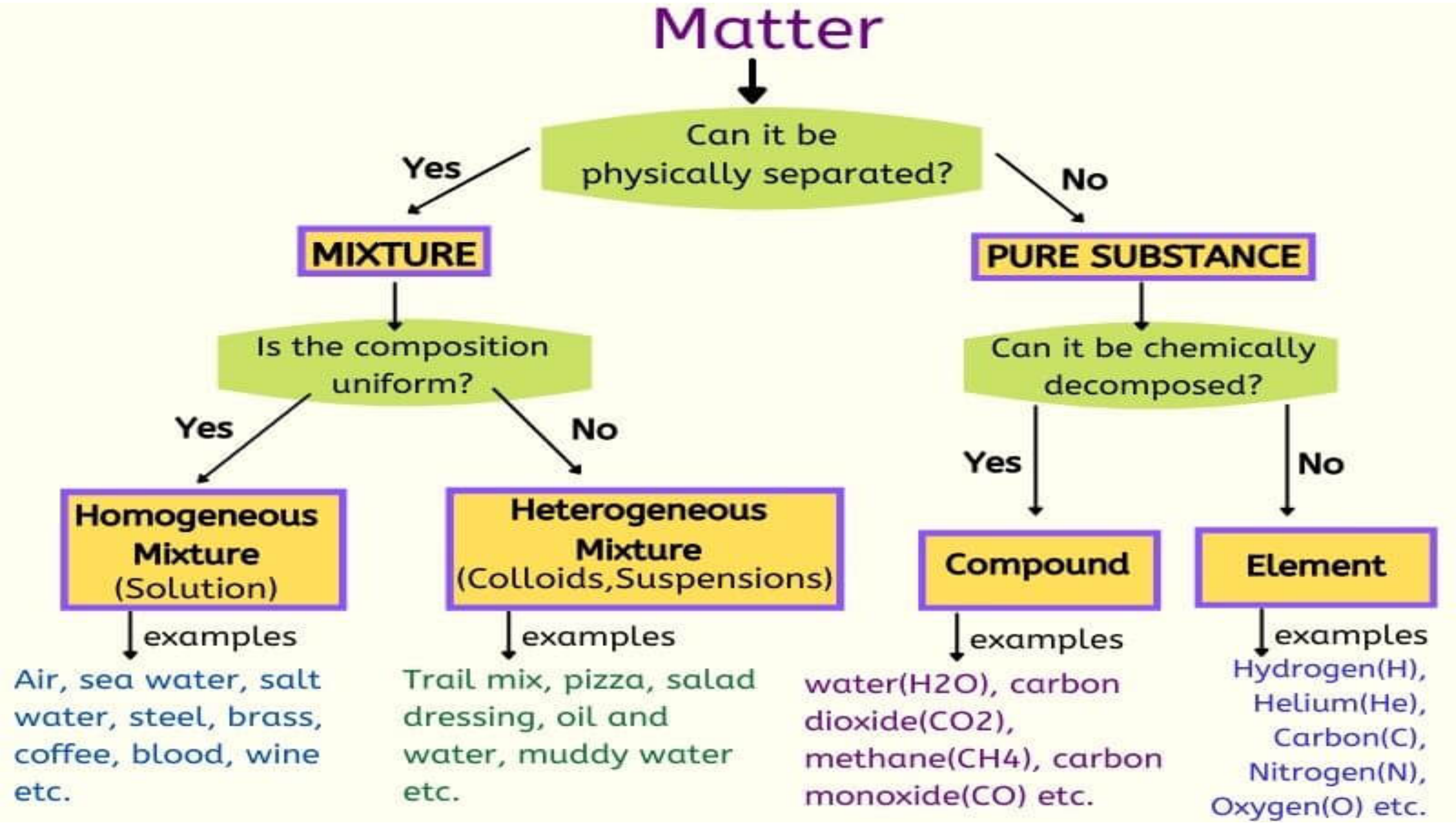
- A mixture of liquids is heated from below and boils.
- The water in the mixture evaporates leaving the remainder.

Condenser

Steam condenses in the tube.

Beaker

Pure, distilled water collects in the beaker.


Matter
Can it be physically separated?
Yes
No
MIXTURE
PURE SUBSTANCE
Is the composition uniform?
Can it be chemically decomposed?
Yes
No
Yes
No
Homogeneous Mixture (Solution)
Heterogeneous Mixture (Colloids,Suspensions)
Compound
Element
examples
Air, sea water, salt water, steel, brass, coffee, blood, wine etc.
examples
Trail mix, pizza, salad dressing, oil and water, muddy water etc.
examples
water(H2O), carbon dioxide(CO2), methane(CH4), carbon monoxide(CO) etc.
examples
Hydrogen(H), Helium(He), Carbon(C), Nitrogen(N), Oxygen(O) etc.